

Pseudocode for an ABM of Brokerage in Matching Markets

1 State, Notation, and Matching Function

Blue notation, equations, and algorithm lines are diagnostics-only measures. They are recorded for $\mathcal{M}$ or downstream figures, exploration scripts, and analyses; they do not enter transition, choice, learning, or matching rules.

For any matrix Z , $Z_{\cdot\ell}$ denotes column ℓ and $Z_{\cdot L}$ denotes the submatrix with ordered column-index set L . For a vector z , z_L denotes the corresponding subvector.

Table 1: Cross-routine symbols and dimensions.

Symbol	Dimension/domain	Definition
<i>Core</i>		
N	scalar	Number of agents, with agent set $[N] = \{1, \dots, N\}$.
ϑ	parameter vector	Simulation parameter vector.
seed	integer	Random seed.
S^t	state tuple	Mutable simulation state at end of period t .
T	integer	Number of simulation periods.
$\mathcal{M} = [m^1; \dots; m^T]$	$\mathbb{R}^{T \times m}$	Measure table; row $m^t = \mathcal{M}_t \in \mathbb{R}^m$.
<i>Parameters</i>		
d	scalar	Agent type dimension.
k_G	scalar	Watts-Strogatz graph degree.
ω	scalar	Satisfaction EWMA weight.
E_{init}	scalar	Initial training steps.
E_{steps}	scalar	Minimum (floor) training steps per period.
η_{lr}	scalar	MLP Adam learning rate.
W_p	scalar	Training-window horizon, in periods.
M	scalar	Per-call training observation cap.
<i>Agent types, networks, and output</i>		
$G^t = (V, \mathcal{E}_G^t)$	graph	Undirected graph, $V = [N] \cup \{N + 1\}$; node $N + 1$ is the broker.
$X^t = [x_1^t, \dots, x_N^t]$	$\mathbb{R}^{d \times N}$	Agent type matrix; $x_i^t = X_{\cdot i}^t \in \mathbb{R}^d$.
γ	$[0, 1] \rightarrow \mathbb{R}^d$	Agent type curve for initial and entrant draws.
q_{ij}^t	scalar	Realized output for match (i, j) .
<i>True matching function</i>		
Q	scalar	Output offset.
ρ	scalar	Quality-interaction weight.
δ	scalar	Regime-gain amplitude.
σ_ε	scalar	Output-noise standard deviation.
c	$\mathbb{R}^d$	Ideal agent type.
A	$\mathbb{R}^{d \times d}$	Symmetric positive definite interaction matrix.
B	$\mathbb{R}^{d \times d}$	Symmetric regime operator.
<i>Calibration and costs</i>		

Symbol	Dimension/domain	Definition
$\bar{q}_{\text{cal}}$	scalar	Calibration mean.
r	scalar	Reservation output.
ϕ	scalar	Broker fee.
c_s	scalar	Self-search cost.
<i>Learning and predictions</i>		
d_b	integer	Broker symmetric pair-feature dimension, $d + d(d + 1)/2$.
$\psi_b(x_i, x_j)$	$\mathbb{R}^{d_b}$	Symmetric broker pair feature vector.
$\mathcal{H}_i^t = (\Xi_i^t, y_i^t, n_i^t)$	$\Xi_i^t \in \mathbb{R}^{d \times n_i^t}, y_i^t \in \mathbb{R}^{n_i^t}$	Agent i history; observation ℓ is partner agent type $\Xi_{i,\ell}^t$ and output $y_{i\ell}^t$.
$\mathcal{H}_b^t = (X_{b,1}^t, X_{b,2}^t, y_b^t, n_b^t)$	$X_{b,1}^t, X_{b,2}^t \in \mathbb{R}^{d \times n_b^t}, y_b^t \in \mathbb{R}^{n_b^t}$	Broker history; observation ℓ stores the two party types and output $y_{b\ell}^t$.
Θ_i^t	NN parameters	Agent i MLP weights.
Θ_b^t	NN parameters	Broker MLP weights.
$v_b^t(i, j)$	scalar	Symmetric broker prediction for unordered pair $\{i, j\}$.
$w_{j \leftarrow i}^t$	scalar	Receiver acceptance score.
$\bar{q}_i^t$	vector	Agent i 's partner-indexed empirical means.
C_i^t	vector	Agent i 's partner-indexed counts.
<i>Period state and market objects</i>		
s_i^t	$\mathbb{R}^2$	Satisfaction scores for self and broker channels.
χ_i^t	binary	Indicator that agent i has ever used the broker.
M_i^t	set	Agent i current-period counterparties.
z_i^t	channel	Channel chosen by demander i .
d_i^t	integer	Current-period demand.
D^t	subset of $[N]$	Current broker clients, $\{i : d_i^t > 0, z_i^t = \text{broker}\}$.
$\mathcal{A}_b^t$	subset of $[N]$	Broker access set, $\text{Roster}^t \cup D^t$.
Roster^t	subset of $[N]$	Standing roster.
rep_b^t	scalar	Broker reputation.
D_+^t	subset of $[N]$	Broker clients with positive residual offer quota.
$\mathcal{D}^t$	list	Demand records (i, z_i^t, d_i^t) with $d_i^t > 0$.
$\mathcal{R}^t$	list	Accepted records $(i, j, z, q, \hat{q}, o_1, o_2)$.

The mutable simulation state at the end of period t is

$$S^t = \left(X^t, G^t, \gamma, \{\mathcal{H}_i^t, \Theta_i^t, \bar{q}_i^t, C_i^t, s_i^t, \chi_i^t, M_i^t\}_{i=1}^N, \mathcal{H}_b^t, \Theta_b^t, \text{Roster}^t, D^t, \text{rep}_b^t \right).$$

The matching function is

$$f(x_i, x_j) = \rho \frac{x_i^\top c + x_j^\top c}{2} + (1 - \rho) g(x_i, x_j) x_i^\top A x_j, \quad g(x_i, x_j) = 1 + \delta \text{sign}(x_i^\top B x_j).$$

2 Top-Level Simulation

Algorithm 1 Brokerage Simulation

Require: Parameters ϑ and seed.

Ensure: Final state S^T and **measure table** $\mathcal{M} \in \mathbb{R}^{T \times m}$.

- 1: $S^0 \leftarrow \text{INITIALIZE}(\vartheta, \text{seed})$
- 2: **for** $t = 1, \dots, T$ **do**

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3:    $(S^t, \mathcal{R}^t) \leftarrow \text{PERIODUPDATE}(S^{t-1}, t, \vartheta)$ 
4:    $\mathcal{M}_t \leftarrow \text{COLLECTMEASURES}(S^t, \mathcal{R}^t, \vartheta)$ 
5: return  $(S^T, \mathcal{M})$ 

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3 Training

For any one-hidden-layer predictor with input dimension p_{in} and hidden width h ,

$$\text{NN}_{\Theta}(\xi) = w_2^{\top} \text{ReLU}(W_1 \xi + b_1) + b_2, \quad \Theta = (W_1, b_1, w_2, b_2),$$

where $\xi \in \mathbb{R}^{p_{\text{in}}}$, $W_1 \in \mathbb{R}^{h \times p_{\text{in}}}$, $b_1, w_2 \in \mathbb{R}^h$, and $b_2 \in \mathbb{R}$.

$$d_b = d + \frac{d(d+1)}{2}, \quad \psi_b(x_i, x_j) = \left[x_i + x_j; \text{vech}\left(\frac{x_i x_j^{\top} + x_j x_i^{\top}}{2}\right) \right].$$

Agent networks use $(p_{\text{in}}, h) = (d, 2d)$; the broker network uses $(p_{\text{in}}, h) = (d_b, 8d)$.

$$\psi_b(x_i, x_j) = \psi_b(x_j, x_i)$$

so broker histories produce one training column per observed unordered pair.

Local variables for training

Symbol	Dimension/domain	Definition
p_{in}, h	integers	Predictor input dimension and hidden width.
ξ	$\mathbb{R}^{p_{\text{in}}}$	Predictor input vector.
Θ	NN parameters	Generic MLP parameters (W_1, b_1, w_2, b_2) .
W_1, b_1, w_2, b_2	matrix, vectors, scalar	One-hidden-layer MLP weights and biases.
d_b	integer	Broker symmetric pair-feature dimension.
ψ_b	function	Symmetric broker pair-feature map.
$\Xi_{\text{tr}}, y_{\text{tr}}$	$\mathbb{R}^{p_{\text{in}} \times n}, \mathbb{R}^n$	Generic training inputs and targets.
$\mathcal{L}$	scalar	Full-batch MSE objective.
e	integer	Update-step index.
(m, v, τ)	Adam state	Persistent Adam moments and step counter; A_i, A_b are the agent/broker instances.
$\beta_1, \beta_2, \epsilon$	scalars	Adam constants $(0.9, 0.999, 10^{-8})$.
n_i, n_b	integers	Current values of history lengths n_i^t and n_b^t .
L_i, L_b	ordered index sets	Training-window indices: observations from the most recent W_p periods, subsampled evenly to at most M .
Δ_i, Δ_b	integers	New observations since last training call.
E_i, E_b	integers	Agent and broker update-step counts.
$\Xi_{i,\text{tr}}, y_{i,\text{tr}}$	$\mathbb{R}^{d \times L_i }, \mathbb{R}^{ L_i }$	Agent i training view of $\mathcal{H}_i^t$.
$\Psi_{b,\text{tr}}, y_{b,\text{tr}}$	$\mathbb{R}^{d_b \times L_b }, \mathbb{R}^{ L_b }$	Broker symmetric-feature training view of $\mathcal{H}_b^t$.

Algorithm 2 TrainNN

Require: Network parameters Θ , inputs $\Xi_{\text{tr}} \in \mathbb{R}^{p_{\text{in}} \times n}$, targets $y_{\text{tr}} \in \mathbb{R}^n$, steps E , learning rate η_{lr} , persistent Adam state (m, v, τ) .

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1: for  $e = 1, \dots, E$  do
2:    $g \leftarrow \nabla_{\Theta} \mathcal{L}(\Theta), \quad \mathcal{L}(\Theta) = n^{-1} \sum_{\ell=1}^n (\text{NN}_{\Theta}(\Xi_{\text{tr},\ell}) - y_{\text{tr},\ell})^2$ 

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3:    $\tau \leftarrow \tau + 1$ 
4:    $m \leftarrow \beta_1 m + (1 - \beta_1)g, \quad v \leftarrow \beta_2 v + (1 - \beta_2)g \odot g$ 
5:    $\Theta \leftarrow \Theta - \eta_{lr} \frac{m/(1 - \beta_1^\tau)}{\sqrt{v/(1 - \beta_2^\tau)} + \epsilon}$ 
6: return  $\Theta, (m, v, \tau)$ 

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Within period-level routines, state variables without a superscript refer to their current-period values.

Algorithm 3 TrainPredictors

Require: State S , period t , parameters $(E_{\text{init}}, E_{\text{steps}}, \eta_{lr}, W_p, M)$.

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1: for  $i = 1, \dots, N$  in parallel do
2:    $n_i \leftarrow n_i^t, \Delta_i \leftarrow$  observations added to  $\mathcal{H}_i$  since its last training call.
3:   if  $n_i > 0, \Delta_i > 0$ , and  $i \bmod 2 = t \bmod 2$  then
4:      $L_i \leftarrow$  indices of  $\mathcal{H}_i$  observations from the most recent  $W_p$  periods; if  $|L_i| > M$ , keep  $M$ 
       spaced evenly across  $L_i$ .
5:      $\Xi_{i,\text{tr}} \leftarrow (\Xi_i^t)_{\cdot L_i}, y_{i,\text{tr}} \leftarrow (y_i^t)_{L_i}$ .
6:      $E_i \leftarrow \max\{E_{\text{steps}}, \lceil E_{\text{init}} \Delta_i / n_i \rceil\}$ .
7:      $(\Theta_i, A_i) \leftarrow \text{TRAINNN}(\Theta_i, \Xi_{i,\text{tr}}, y_{i,\text{tr}}, E_i, \eta_{lr}, A_i); \Delta_i \leftarrow 0$ .
8:    $n_b \leftarrow n_b^t, \Delta_b \leftarrow$  broker observations since last training.
9:   if  $n_b > 0$  and  $\Delta_b > 0$  then
10:     $L_b \leftarrow$  indices of  $\mathcal{H}_b$  observations from the most recent  $W_p$  periods; if  $|L_b| > M$ , keep  $M$ 
       spaced evenly across  $L_b$ .
11:     $\Psi_{b,\text{tr}} \leftarrow \left[ \psi_b \left( (X_{b,1}^t)_{\cdot \ell}, (X_{b,2}^t)_{\cdot \ell} \right) \right]_{\ell \in L_b}$ .
12:     $y_{b,\text{tr}} \leftarrow (y_b^t)_{L_b}$ .
13:     $E_b \leftarrow \max\{E_{\text{steps}}, \lceil E_{\text{init}} \Delta_b / n_b \rceil\}$ .
14:     $(\Theta_b, A_b) \leftarrow \text{TRAINNN}(\Theta_b, \Psi_{b,\text{tr}}, y_{b,\text{tr}}, E_b, \eta_{lr}, A_b); \Delta_b \leftarrow 0$ .

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4 Initialization

Calibration draws random pairs (i_ℓ, j_ℓ) and sets

$$\bar{q}_{\text{cal}} = \frac{1}{10,000} \sum_{\ell=1}^{10,000} \{Q + f(x_{i_\ell}, x_{j_\ell})\},$$

$$r = f_r \bar{q}_{\text{cal}}, \quad \phi = c_s = \lambda_c \bar{q}_{\text{cal}}.$$

Local variables for Initialize

Symbol	Dimension/domain	Definition
λ_c	scalar	Cost rate used to set ϕ and c_s .
f_r	scalar	Reservation coefficient; sets $r = f_r \bar{q}_{\text{cal}}$ (default 0.60).
(i_ℓ, j_ℓ)	pair in $[N]^2$	Random calibration pair.
d_γ	integer	Number of active curve dimensions.
p_{rewire}	scalar	Watts-Strogatz rewiring probability.
ν_ℓ	integer	Frequency for active curve coordinate ℓ .

Local variables for Initialize (continued)

Symbol	Dimension/domain	Definition
ψ_ℓ	scalar	Phase for active curve coordinate ℓ .
A_0	$\mathbb{R}^{d \times d}$	Raw matrix used to construct A .
Σ_x	$\mathbb{R}^{d \times d}$	Empirical agent type second moment.
B_0	$\mathbb{R}^{d \times d}$	Raw zero-trace symmetric regime matrix.
B_{raw}	$\mathbb{R}^{d \times d}$	Orthogonalized pre-normalization regime matrix.
$\Psi_{b,\text{seed}}, y_{b,\text{seed}}$	$\mathbb{R}^{d_b \times n_b^0}, \mathbb{R}^{n_b^0}$	Broker seed training batch using symmetric pair features.

Local variables for InitializeAgent

Symbol	Dimension/domain	Definition
σ_x	scalar	Agent type noise scale.
u_i	scalar	Agent i curve position.
ϵ_i	$\mathbb{R}^d$	Agent type noise for agent i .
π_{ij}^{ent}	scalar	Entrant attachment probability.
$\mathcal{V}_i^{\text{ent}}$	subset of $[N] \setminus \{i\}$	Entrant attachment set.

Algorithm 4 InitializeAgent

Require: State S , agent slot i , period counter t , parameters ϑ .

Ensure: Agent slot i is initialized in S .

- 1: Draw $u_i \sim \text{Unif}[0, 1]$ and $\epsilon_i \sim \mathcal{N}(0, \sigma_x^2 I_d/d)$.
 - 2: $x_i^t \leftarrow (\gamma(u_i) + \epsilon_i) / \|\gamma(u_i) + \epsilon_i\|_2$.
 - 3: Reset $\mathcal{H}_i^t = (\Xi_i^t, y_i^t, n_i^t)$ to zero observations, $\bar{q}_i^t$, C_i^t , χ_i^t , and M_i^t ; draw fresh agent MLP weights Θ_i^t with output bias Q .
 - 4: **if** $t = 0$ **then**
 - 5: $s_i^t \leftarrow (0, 0)$.
 - 6: **else**
 - 7: Remove i from Roster^t , D^t , and M_j^t for all j ; delete all incident edges in G^t ; set $\bar{q}_{ji}^t, C_{ji}^t \leftarrow 0$ for all $j \neq i$.
 - 8: Sample $\mathcal{V}_i^{\text{ent}} \subset [N] \setminus \{i\}$ without replacement, $|\mathcal{V}_i^{\text{ent}}| = \min(\lfloor k_G/2 \rfloor, N - 1)$, with

$$\pi_{ij}^{\text{ent}} \propto \exp\{-\|x_i^t - x_j^t\|_2^2\}.$$
 - 9: Add edges (i, j) to G^t for all $j \in \mathcal{V}_i^{\text{ent}}$.
 - 10: $s_{i,\text{self}}^t \leftarrow \begin{cases} |\mathcal{V}_i^{\text{ent}}|^{-1} \sum_{j \in \mathcal{V}_i^{\text{ent}}} s_{j,\text{self}}^t, & \mathcal{V}_i^{\text{ent}} \neq \emptyset, \\ 0, & \mathcal{V}_i^{\text{ent}} = \emptyset. \end{cases}$
 - 11: $s_{i,\text{broker}}^t \leftarrow \text{rep}_b^t$.
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Algorithm 5 Initialize

Require: Parameters ϑ and seed.

Ensure: Initial state S^0 .

- 1: Draw curve frequencies $\nu_\ell \sim \text{Unif}\{1, \dots, 5\}$ and phases $\psi_\ell \sim \text{Unif}[0, 2\pi)$ for $\ell = 1, \dots, d_\gamma$, defining γ .
- 2: Allocate empty agent and broker containers in S^0 and store γ .

- 3: **for** $i = 1, \dots, N$ **do**
- 4: INITIALIZEAGENT($S^0, i, 0, \vartheta$).
- 5: Draw ideal agent type c as a noisy curve point.
- 6: Draw $A_0 \in \mathbb{R}^{d \times d}$ iid standard normal and set $A \leftarrow A_0^\top A_0 \cdot d / \text{tr}(A_0^\top A_0)$.
- 7: $\Sigma_x \leftarrow N^{-1} \sum_i x_i^0 (x_i^0)^\top$.
- 8: Draw symmetric zero-trace B_0 and set

$$B_{\text{raw}} \leftarrow B_0 - \frac{\text{tr}(\Sigma_x B_0 \Sigma_x A)}{\text{tr}(\Sigma_x A \Sigma_x A)} A, \quad B \leftarrow B_{\text{raw}} / \|B_{\text{raw}}\|_F.$$

- 9: Calibrate $(\bar{q}_{\text{cal}}, r, \phi, c_s)$.
- 10: Build G^0 as a Watts-Strogatz graph on $[N]$ with degree k_G and rewiring probability p_{rewire} ; add isolated broker node $N + 1$.
- 11: Initialize $\mathcal{H}_b^0$, Θ_b^0 , and rep_b^0 .
- 12: Sample $\text{Roster}^0 \subset [N]$ uniformly with $|\text{Roster}^0| = \lceil 0.2N \rceil$; add broker edges $(i, N + 1)$ for all roster members.
- 13: **for** each unordered non-broker edge $\{i, j\} \in \mathcal{E}_G^0$ **do**
- 14: Draw q_{ij} once; append (x_j^0, q_{ij}) to $\mathcal{H}_i^0$ and (x_i^0, q_{ij}) to $\mathcal{H}_j^0$; update $\bar{q}_{ij}^0, C_{ij}^0, \bar{q}_{ji}^0, C_{ji}^0$.
- 15: Let $\mathcal{B}_{\text{seed}} \leftarrow \{\{i, j\} \in \mathcal{E}_G^0 : i, j \in \text{Roster}^0\}$; shuffle and keep at most 100 edges.
- 16: **for** each $\{i, j\} \in \mathcal{B}_{\text{seed}}$ **do**
- 17: Append (x_i^0, x_j^0, q_{ij}) to $\mathcal{H}_b^0$; increment n_b^0 .
- 18: **if** $n_b^0 > 0$ **then**
- 19: Set rep_b^0 to mean broker seed output.
- 20: **else**
- 21: Set $\text{rep}_b^0 \leftarrow 0$.
- 22: Set $s_{i, \text{self}}^0$ to mean own seed output and $s_{i, \text{broker}}^0 \leftarrow \text{rep}_b^0$.
- 23: **for** each i with $n_i^0 > 0$ **do**
- 24: $\Theta_i^0 \leftarrow \text{TRAINNN}(\Theta_i^0, \Xi_i^0, y_i^0, E_{\text{init}}, \eta_{\text{lr}})$.
- 25: **if** $n_b^0 > 0$ **then**
- 26: $\Psi_{b, \text{seed}} \leftarrow \left[\psi_b \left((X_{b,1}^0)_{\cdot \ell}, (X_{b,2}^0)_{\cdot \ell} \right) \right]_{\ell=1}^{n_b^0}$, $y_{b, \text{seed}} \leftarrow y_b^0$.
- 27: $\Theta_b^0 \leftarrow \text{TRAINNN}(\Theta_b^0, \Psi_{b, \text{seed}}, y_{b, \text{seed}}, E_{\text{init}}, \eta_{\text{lr}})$.
- 28: **return** S^0 .

5 Channel Choice

Local variables for ChooseChannel

Symbol	Dimension/domain	Definition
U_{self}	scalar	Self-channel score.
U_{broker}	scalar	Broker-channel score.

Algorithm 6 ChooseChannel

Require: Agent i state (s_i, χ_i) , broker reputation rep_b , RNG.

- 1: $U_{\text{self}} \leftarrow s_{i, \text{self}}$.
- 2: $U_{\text{broker}} \leftarrow \chi_i s_{i, \text{broker}} + (1 - \chi_i) \text{rep}_b$.

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3: if  $U_{\text{self}} > U_{\text{broker}}$  then
4:   return self.
5: if  $U_{\text{broker}} > U_{\text{self}}$  then
6:   return broker.
7: return self w.p.  $1/2$ , else broker

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$\triangleright U_{\text{self}} = U_{\text{broker}}.$

6 One Period

Local variables for PeriodUpdate

Symbol	Dimension/domain	Definition
K	integer	Active-demand bound.
p_{demand}	scalar	Per-position demand probability.
p_{roster}	scalar	Roster-churn probability.
η_{exit}	scalar	Agent exit probability.
Δ_{net}	integer	Network-measure interval.

Algorithm 7 PeriodUpdate

Require: Previous state S^{t-1} , period t , parameters ϑ .
Ensure: Updated state S^t and accepted records $\mathcal{R}^t$.

- 1: Clear accumulators, set $M_i^t \leftarrow \emptyset$ for all i , and set $D^t \leftarrow \emptyset$.
- 2: Refresh roster: remove each roster member with probability p_{roster} , then replenish uniformly to size $\lceil 0.2N \rceil$.
- 3: $\mathcal{E}_G^t \leftarrow (\mathcal{E}_G^t \setminus \{\{i, N+1\} : i \in [N]\}) \cup \{\{i, N+1\} : i \in \text{Roster}^t\}$.
- 4: $\mathcal{D}^t \leftarrow \emptyset$.
- 5: **for** $i = 1, \dots, N$ **do**
- 6: Draw $d_i^t \sim \text{Binomial}(K, p_{\text{demand}})$.
- 7: **if** $d_i^t > 0$ **then**
- 8: $z_i^t \leftarrow \text{CHOOSECHANNEL}(i, \text{rep}_b^{t-1})$.
- 9: Append (i, z_i^t, d_i^t) to $\mathcal{D}^t$.
- 10: **if** $z_i^t = \text{broker}$ **then**
- 11: Add i to D^t .
- 12: $\mathcal{E}_G^t \leftarrow (\mathcal{E}_G^t \setminus \{\{i, N+1\} : i \in [N]\}) \cup \{\{i, N+1\} : i \in \text{Roster}^t \cup D^t\}$.
- 13: $\text{TRAINPREDICTORS}(S, t, \vartheta)$.
- 14: $\mathcal{R}^t \leftarrow \text{OFFERMARKET}(S, \mathcal{D}^t, \vartheta)$.
- 15: Synchronize broker-node edges to Roster^t , D^t , and current broker-channel matches.
- 16: $\text{UPDATESATISFACTION}(S, \mathcal{D}^t, \mathcal{R}^t)$.
- 17: **if** $D^t \neq \emptyset$ **then**
- 18: $\text{rep}_b^t \leftarrow |D^t|^{-1} \sum_{i \in D^t} s_{i, \text{broker}}^t$.
- 19: **else**
- 20: $\text{rep}_b^t \leftarrow \text{rep}_b^{t-1}$.
- 21: **if** $t \bmod \Delta_{\text{net}} = 0$ **then**
- 22: Recompute broker betweenness, Burt constraint, and effective size on the post-matching, pre-turnover graph G^t .
- 23: $Y^t \leftarrow \text{COLLECTMEASURES}(S^t, \mathcal{R}^t, \vartheta)$ on the post-matching, pre-turnover state.
- 24: **for** $i = 1, \dots, N$ **do**

25: **if** agent i exits with probability η_{exit} **then**
26: INITIALIZEAGENT(S^t, i, t, ϑ).
27: Synchronize broker-node edges to the post-turnover roster, clients, and active matches.
28: **return** ($S^t, \mathcal{R}^t, Y^t$).

7 Offer Market

Local variables for OfferMarket

Symbol	Dimension/domain	Definition
$\hat{q}_i^t$	$\mathbb{R}^d \rightarrow \mathbb{R}$	Agent predictor.
$\hat{q}_b^t$	$\mathbb{R}^{d_b} \rightarrow \mathbb{R}$	Broker predictor.
$\mathcal{N}_G^t(i)$	subset of $[N]$	Non-broker graph neighbors of i .
U_i^t	subset of $[N]$	Self-search stranger pool independently sampled for demander i .
$n_{\text{strangers}}$	integer	Per-demander stranger-pool size.
$v_{i \rightarrow j}^{\text{self}, t}$	scalar	Self-offer score.
Ω_i	integer	Residual offer quota for demander i .
Ω	$\mathbb{Z}_{\geq 0}^N$	Residual quota vector.
O^t	map	Directed offer book.
O_{ij}^t	offer record	Record (i, j, z, v) for offer $i \rightarrow j$.
$\mathcal{P}_O^t$	set	Unordered pairs represented in O^t .
$\mathcal{U}_i^{\text{self}}$	subset of $[N]$	Self-search candidate set.
$\mathcal{I}_i^{\text{self}}$	list	Self-search candidates sorted by score.
$\mathcal{P}_b^t$	set	Feasible unordered broker-pair set.
$\mathcal{I}_{b,i}$	list	Accessible broker candidates for client i , sorted by v_b^t .
$\mathcal{B}_b^t$	list	Selected directed broker offers.
o, o', o_{ij}, o_{ji}	offer records	Offer records; fields include $o.z$ and $o.from$.
z^*	channel	Accepted-record channel.
α, β	indices	Ordered endpoints induced by one unordered broker pair.
ι	$\{0, 1\}$	Return value of AddOffer.

The symmetric broker score is

$$v_b^t(i, j) = \hat{q}_b^t(\psi_b(x_i^t, x_j^t)).$$

Let $\mathcal{N}_G^t(i) = \{j \in [N] : (i, j) \in \mathcal{E}_G^t\}$ exclude the broker node. The self-search proposal score is defined only for observed graph neighbors and the demander-specific stranger pool:

$$v_{i \rightarrow j}^{\text{self}, t} = \begin{cases} \bar{q}_{ij}^t, & j \in \mathcal{N}_G^t(i), C_{ij}^t > 0, \\ \hat{q}_i^t(x_j^t), & j \in U_i^t \setminus \mathcal{N}_G^t(i), \\ \perp, & \text{otherwise.} \end{cases}$$

Thus direct pair means dominate predictor extrapolations when pair-specific experience exists; $\perp$ denotes no valid self-search score. The receiver-side acceptance score for a one-sided offer is

$$w_{j \leftarrow i}^t = \begin{cases} \bar{q}_{ji}^t, & (i, j) \in \mathcal{E}_G^t, C_{ji}^t > 0, \\ \hat{q}_j^t(x_i^t), & \text{otherwise.} \end{cases}$$

The market creates at most d_i^t outgoing offers from demander i . Receivers have no per-period capacity constraint; acceptance is pairwise, so a receiver may accept multiple offers in the same period. A demanded position can remain unmatched if no acceptable offer is formed.

Algorithm 8 AddOffer

Require: Offer book O^t , unordered-pair set $\mathcal{P}_O^t$, offer (i, j, z, v) .

- 1: **if** O_{ij}^t is defined **then**
- 2: **return** 0.
- 3: $O_{ij}^t \leftarrow (i, j, z, v)$.
- 4: **if** O_{ji}^t is undefined **then**
- 5: $\mathcal{P}_O^t \leftarrow \mathcal{P}_O^t \cup \{\{i, j\}\}$.
- 6: **return** 1.

Algorithm 9 OfferMarket

Require: State S , demand records $\mathcal{D}^t$, parameters ϑ .
Ensure: Accepted records $\mathcal{R}^t$.

- 1: For all $(i, z_i, d_i) \in \mathcal{D}^t$, set residual offer quota $\Omega_i \leftarrow d_i$.
- 2: $O^t \leftarrow \emptyset$, $\mathcal{P}_O^t \leftarrow \emptyset$, $\mathcal{R}^t \leftarrow \emptyset$.
- 3: **for** each $(i, \text{self}, d_i) \in \mathcal{D}^t$ with $\Omega_i > 0$ **do**
- 4: Draw $U_i^t \subset [N]$ uniformly without replacement, $|U_i^t| = \min(n_{\text{strangers}}, N)$.
- 5: $\mathcal{U}_i^{\text{self}} \leftarrow \{j \in \mathcal{N}_G^t(i) : C_{ij}^t > 0\} \cup (U_i^t \setminus \mathcal{N}_G^t(i))$.
- 6: Remove from $\mathcal{U}_i^{\text{self}}$ the elements of $\{i\} \cup M_i^t$.
- 7: $\mathcal{I}_i^{\text{self}} \leftarrow \text{argsort}_{j \in \mathcal{U}_i^{\text{self}}} : v_{i \rightarrow j}^{\text{self}, t} > r (-v_{i \rightarrow j}^{\text{self}, t})$.
- 8: **for** the first $\min(\Omega_i, |\mathcal{I}_i^{\text{self}}|)$ agents j in $\mathcal{I}_i^{\text{self}}$ **do**
- 9: ADDOFFER($O^t, \mathcal{P}_O^t, i, j, \text{self}, v_{i \rightarrow j}^{\text{self}, t}$).
- 10: $\mathcal{A}_b^t \leftarrow \text{Roster}^t \cup D^t$.
- 11: $D_+^t \leftarrow \{i : (i, \text{broker}, d_i) \in \mathcal{D}^t, \Omega_i > 0\}$.
- 12: $\mathcal{P}_b^t \leftarrow \{\{i, j\} : i \neq j, (i \in D_+^t, j \in \mathcal{A}_b^t) \text{ or } (j \in D_+^t, i \in \mathcal{A}_b^t)\}$.
- 13: $\mathcal{B}_b^t \leftarrow \emptyset$.
- 14: **for** each $i \in D_+^t$ **do**
- 15: $\mathcal{I}_{b,i} \leftarrow \text{argsort}_{j \in \mathcal{A}_b^t : \{i, j\} \in \mathcal{P}_b^t, v_b^t(i, j) > r (-v_b^t(i, j))}$.
- 16: **for** the first $\min(\Omega_i, |\mathcal{I}_{b,i}|)$ agents j in $\mathcal{I}_{b,i}$ **do**
- 17: Append $(i, j, v_b^t(i, j))$ to $\mathcal{B}_b^t$.
- 18: Sort $\mathcal{B}_b^t$ by descending v_b^t with the deterministic unordered-pair tie key.
- 19: **for** each $(i, j, v) \in \mathcal{B}_b^t$ **do**
- 20: **if** $\Omega_i > 0$ **then**
- 21: $\iota \leftarrow \text{ADDOFFER}(O^t, \mathcal{P}_O^t, i, j, \text{broker}, v)$.
- 22: $\Omega_i \leftarrow \Omega_i - \iota$.
- 23: **for** each $\{i, j\} \in \mathcal{P}_O^t$ **do**
- 24: $o_{ij} \leftarrow O_{ij}^t$ if defined, else $\emptyset$; $o_{ji} \leftarrow O_{ji}^t$ if defined, else $\emptyset$.
- 25: **if** $o_{ij} \neq \emptyset$ and $o_{ji} \neq \emptyset$ **then**
- 26: FINALIZEMATCH($S, \mathcal{R}^t, o_{ij}, o_{ji}$).
- 27: **else if** $o_{ij} \neq \emptyset$ and $w_{j \leftarrow i}^t > r$ **then**
- 28: FINALIZEMATCH($S, \mathcal{R}^t, o_{ij}, \emptyset$).
- 29: **else if** $o_{ji} \neq \emptyset$ and $w_{i \leftarrow j}^t > r$ **then**
- 30: FINALIZEMATCH($S, \mathcal{R}^t, \emptyset, o_{ji}$).

31: **return** $\mathcal{R}^t$.

Algorithm 10 FinalizeMatch

Require: State S , accepted-record list $\mathcal{R}^t$, offer $o = (i, j, z, v)$, optional reverse offer o' .

```

1: if  $j \in M_i^t$  then
2:   return .
3: Draw  $q_{ij}^t = Q + f(x_i^t, x_j^t) + \varepsilon$ ,  $\varepsilon \sim \mathcal{N}(0, \sigma_\varepsilon^2)$ .
4: Append columns  $x_j^t$  to  $\Xi_i$  and  $x_i^t$  to  $\Xi_j$ ; append  $q_{ij}^t$  to  $y_i$  and  $y_j$ ; increment  $n_i$  and  $n_j$ .
5: Update  $\bar{q}_{ij}$ ,  $C_{ij}$  and  $\bar{q}_{ji}$ ,  $C_{ji}$ .
6: Add edge  $(i, j)$  to  $G^t$ ; add  $j$  to  $M_i^t$  and  $i$  to  $M_j^t$ .
7: if  $z = \text{broker}$  or  $(o' \neq \emptyset \text{ and } o'.z = \text{broker})$  then
8:   Append column  $[x_i^t; x_j^t]$  to  $\Xi_b$ ; append  $q_{ij}^t$  to  $y_b$ ; increment  $n_b$ .
9:    $z^* \leftarrow \text{broker}$ .
10: else
11:    $z^* \leftarrow \text{self}$ .
12: Append accepted record  $(i, j, z^*, q_{ij}^t, v, o, o')$  to  $\mathcal{R}^t$ .

```

8 Satisfaction and Measures

Local variables for satisfaction

Symbol	Dimension/domain	Definition
Y_i	scalar	Agent i period satisfaction numerator.
$n_{i,b}$	integer	Accepted broker offers credited to demander i .
$\tilde{q}_i^t$	scalar	Channel-specific satisfaction input.
ζ	accepted record	Iterator over $\mathcal{R}^t$.
$\mathbf{1}\{\cdot\}$	indicator	Equals 1 when its condition is true and 0 otherwise.

Algorithm 11 UpdateSatisfaction

Require: State S , demand records $\mathcal{D}^t$, accepted records $\mathcal{R}^t$.

```

1: for each  $(i, z_i, d_i) \in \mathcal{D}^t$  do
2:    $Y_i \leftarrow \sum_{\zeta \in \mathcal{R}^t} \sum_{o \in \{\zeta.o_1, \zeta.o_2\}} \mathbf{1}\{o \neq \emptyset, o.\text{from} = i\} \zeta.q$ .
3:    $n_{i,b} \leftarrow \sum_{\zeta \in \mathcal{R}^t} \sum_{o \in \{\zeta.o_1, \zeta.o_2\}} \mathbf{1}\{o \neq \emptyset, o.\text{from} = i, o.z = \text{broker}\}$ .
4:   if  $z_i = \text{self}$  then
5:      $\tilde{q}_i^t \leftarrow Y_i/d_i - c_s$ .
6:   else
7:      $\tilde{q}_i^t \leftarrow (Y_i - \phi n_{i,b})/d_i$ ;  $\chi_i^t \leftarrow 1$ .
8:    $s_{i,z_i}^t \leftarrow (1 - \omega)s_{i,z_i}^{t-1} + \omega \tilde{q}_i^t$ .

```

CollectMeasures. $\text{COLLECTMEASURES}(S^t, \mathcal{R}^t, \vartheta)$ is a pure measure-collection function called after matching and before entry/exit turnover. It records demand, outsourcing, accepted matches, realized output, selected-sample prediction quality, deterministic holdout prediction quality, broker access and assessment counts, satisfaction, reputation, roster size, broker access size, agent-node degree summaries excluding the broker node, and cached broker network measures. The measures themselves do not feed back into S^{t+1} .